Amortized Bayesian parameter recovery for dynamical systems with conditional flow matching

The `amortix` package: architecture, evaluation methodology, and benchmarks

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Abstract

We describe `amortix`, a package for amortized Bayesian estimation of the parameters of dynamical systems — ordinary and stochastic differential equations and discretized PDEs — from noisy observations at a finite set of time points. A conditional flow-matching model is trained once on simulated (parameter, data) pairs and thereafter produces posterior samples for any new dataset without likelihood evaluations or per-dataset optimization; a single trained network handles any number and placement of observation points, the setting required for sequential experiment refinement and Bayesian experimental design. The package couples three components that we argue are only useful together: (i) a set-transformer encoder whose input representation is learnable rather than prescribed — a monotone mixture-of-scales reparametrization of the value axis, attention phases computed from physical observation times, and explicit set-cardinality features; (ii) a benchmark suite of fifteen systems spanning finance, neuroscience, epidemiology, ecology, pharmacokinetics, classical mechanics, and reaction–diffusion PDEs; and (iii) an evaluation harness in which simulation-based calibration is used only as a screen, and the instrument of record is comparison against exact or validated Markov chain Monte Carlo reference posteriors on the same observed points. On the fixed-design suite, 31 of 32 parameters across nine systems pass simulation-based calibration with a mean coverage error of 1.3 percentage points; on the variable-design suite, posterior widths match exact references to within 5–7% uniformly over design sizes $K = 5$–100, and the two residual defects found anywhere in the suite decay as measured power laws of the training budget. We compare against neural posterior estimation and flow-matching posterior estimation from the `sbi` package, both network families of BayesFlow 2, and a port of the Simformer, on three systems with exact or validated references. On a multiplicative process, all external configurations either inflate the volatility posterior by factors of 1.4–3.2 or become overconfident (widths 0.4–0.9) depending on the input coordinates they are given, while the learnable representation recovers the exact posterior from raw data. On an additive control process the same external tools improve markedly — the systematic bias disappears and the width inflation shrinks from 2.5–3.2 to 1.6–1.9 — which localizes the dominant failure to input conditioning while showing that a residual representation gap remains even in the benign case. We document this failure mechanism — the conditioning of the encoder input — in three distinct forms, together with learnable modules that repair each one.

1 Introduction

Estimating the parameters of a differential-equation model from noisy, irregularly sampled trajectory data is among the most common inverse problems in applied science. Two features make the fully Bayesian version of this problem expensive. First, for most stochastic and partially observed systems, the likelihood $p(\mathcal{D} \mid m)$ requires integrating over latent paths between observations and is intractable or costly. Second, modern workflows evaluate posteriors *repeatedly*: calibration studies process thousands of synthetic datasets; sequential experiments update the posterior after every new measurement; Bayesian experimental design compares candidate observation schedules by the posteriors they would produce. Per-dataset MCMC or optimization inside such loops is the dominant cost even when a likelihood is available, and is impossible when it is not.

Simulation-based inference (SBI) addresses both issues by learning a conditional generative model $q_\theta(m \mid \mathcal{D})$ from simulated pairs $(m_j, \mathcal{D}_j)$, amortizing the cost of inference into a one-time training phase (Cranmer et al., 2020). Neural posterior estimation (NPE) and its sequential variants (Papamakarios and Murray, 2016; Greenberg et al., 2019) fit normalizing flows by maximum likelihood; flow-matching and diffusion variants (Wildberger et al., 2023; Gloeckler et al., 2024) replace the density estimator with transport-based generative models. Mature software exists: the `sbi` toolkit (Boelts et al., 2024), BayesFlow (Radev et al., 2020), and the `sbibm` benchmark collection (Lueckmann et al., 2021).

This report describes `amortix`, a compact PyTorch package (about 3,500 lines; one default model of 4.8×10^5 weights) for amortized Bayesian parameter recovery in dynamical systems. Three aspects of its design distinguish it from the toolkits above, and the report is organized around measuring their consequences.

The first concerns the interface between raw data and the network. Conditional density estimators are conventionally fitted to a fixed-length data vector after global standardization; the estimator sees whatever coordinates the user chose. We show in Sec. 8.5 that on multiplicative and heavy-tailed processes this convention fails in a specific, reproducible way across four different generative engines from three codebases, and that supplying hand-derived coordinates does not repair it uniformly: with identical inputs, different implementations end up over-dispersed, overconfident, or biased. In `amortix` the transformation from observations to network inputs is itself part of the trained model — a monotone mixture-of-scales reparametrization of the value axis, observation times entering attention as continuous phases, and explicit design-size features (Sec. 4); each component is motivated by a measured failure and validated by an ablation.

The second concerns observation designs. `amortix` trains a single network to approximate $p(m \mid S)$ coherently for every subset S of the measurement points of an experiment — arbitrary sizes and placements, including spatio-temporal sensor designs for PDEs (Sec. 8.2). This is the object consumed by sequential refinement and Bayesian experimental design, and supporting it changes the training procedure (Sec. 5): designs must be resampled fresh at every optimizer step, and the distribution of design sizes becomes a modeling choice with measurable consequences.

The third concerns evaluation. Simulation-based calibration (Talts et al., 2018) is a necessary-condition test whose statistical power at practical sample sizes is limited; we measure that limit (Sec. 6) and use SBC only as a screen. The instrument of record throughout is the comparison of posterior mean and width against an exact or validated-MCMC reference posterior on the same observed points, and the package ships this machinery — closed forms, exact-likelihood adaptive Metropolis samplers, chain validation — as part of its public API. The concern motivating this discipline is documented in the SBI literature: learned posteriors can be confidently wrong in ways that rank diagnostics do not detect (Hermans et al., 2022).

The remainder of the report presents the method (Sec. 3), the architecture (Sec. 4), training procedures (Sec. 5), the evaluation methodology (Sec. 6), the benchmark suite (Sec. 7), experimental results including comparisons with `sbi`, BayesFlow, and the Simformer on three systems with exact or validated references (Sec. 8), an analysis of the failure modes encountered during development (Sec. 9), and limitations (Sec. 10). Appendix A defines every benchmark system with its equations, priors, and observation model; Appendix B details the reference posteriors; Appendix C maps the terminology of this report to package options.

2 Problem formulation

Let a dynamical system depend on a parameter vector $m \in \mathbb{R}^d$ with a known prior $p(m)$, uniform on a box in all benchmarks below. The system produces a state trajectory $x(t)$ — through an ODE $\dot{x} = f(x; m)$, an SDE $dx = f(x; m)\,dt + \sigma(x; m)\,dW_t$, or a spatially discretized PDE — and the trajectory is observed through a noisy channel at K points:

$$y_i = h\big(x(t_i)\big) + \varepsilon_i, \qquad \mathcal{D} = \{(t_i, y_i)\}_{i=1}^{K}. \tag{1}$$

The observation operator h may expose only part of the state: in the benchmarks below, only the price of an asset while its stochastic volatility is latent, only the membrane voltage of a neuron while its gating variables are latent, or only three spatial sensors of a PDE field. The target is the posterior $p(m \mid \mathcal{D}) \propto p(\mathcal{D} \mid m)\, p(m)$.

Three standing assumptions delimit the setting. The parameter is finite- and low-dimensional ($d = 2$–5 in every benchmark below) with a prior that factorizes across components and has tractable marginal CDFs (uniform boxes throughout; the construction extends to any such prior). The simulator can be run forward cheaply and repeatedly, but nothing else is required of it — no likelihood evaluations, no gradients. Observation times (and sensor indices) are known exactly; the measurement noise ε_i is independent across observations with a known per-system law.

Running example. The Ornstein–Uhlenbeck process is used throughout to make constructions concrete: $dX = -\theta X\,dt + \sigma\,dW$ with $m = (\theta, \sigma)$, prior box $[0.3, 3] \times [0.2, 1.5]$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$. A dataset is, for instance, $K = 74$ noisy values of X on a fixed two-rate grid, or — in the variable-design setting — any K values at arbitrary times. Its posterior has the features that recur across the suite: the diffusion coefficient σ is identified sharply by high-frequency increments; the reversion rate θ only weakly over a finite horizon; and the width of both marginals must shrink as observations accumulate. An estimator can therefore be wrong in the posterior mean, in the posterior width, or in the width-versus-K trend, and the evaluation methodology of Sec. 6 is built to detect all three.

Amortization. Sampling from the joint distribution is easy — draw $m_j \sim p(m)$, simulate, apply the observation channel — and yields training pairs $(m_j, \mathcal{D}_j)$. A conditional generative model trained on such pairs approximates the posterior for every dataset simultaneously; inference for a new dataset is a forward pass.

Design amortization. We additionally require that a single trained network be correct for any number K and placement of observation times (and sensor positions). Formally, for every subset S of the measurement points of one experiment, the network must approximate $p(m \mid S)$; in particular the posterior must widen consistently as points are removed. This object is what sequential refinement and Bayesian experimental design consume: candidate designs are compared by the posteriors they would produce.

3 Posterior sampling by conditional flow matching

Parameter normalization. Since the prior factorizes with known one-dimensional CDFs, parameters are mapped componentwise through

$$z = \Phi^{-1}\big(F_{\text{prior}}(m)\big) \in \mathbb{R}^d, \tag{2}$$

where Φ is the standard normal CDF. In z-coordinates the prior is exactly $\mathcal{N}(0, I)$ and box constraints disappear. The choice fixes the uninformative limit: when the data carry no information the posterior coincides with the base distribution and the transport below reduces to the identity, so the network need not learn anything to represent ignorance correctly.

Conditional flow matching. The posterior is represented as the endpoint of a learned transport (Lipman et al., 2023). Let $z_0 \sim \mathcal{N}(0, I)$ and let z_1 be the normalized parameter drawn jointly with its dataset $\mathcal{D}$. Define the linear interpolation $z_t = (1-t)z_0 + tz_1$ and train a velocity network by regression:

$$\mathcal{L}(\theta) = \mathbb{E}_{m,\mathcal{D},\ z_0\sim\mathcal{N}(0,I),\ t\sim U[0,1]} \big\| v_\theta(z_t,\ t,\ \mathcal{D}) - (z_1 - z_0) \big\|^2. \tag{3}$$

The minimizer of (3) is the marginal velocity field whose flow transports $\mathcal{N}(0, I)$ at $t = 0$ to $p(z_1 \mid \mathcal{D})$ at $t = 1$ (Lipman et al., 2023). Sampling integrates $\dot{z} = v_\theta(z, t, \mathcal{D})$ with 20 midpoint steps (a finer RK4/60 solver changes no benchmark result; Sec. 8.3) and maps back through the inverse of (2). Evaluation uses an exponential moving average of the weights. The same estimator family, with a different conditioning architecture, is available in `sbi` as FMPE (Wildberger et al., 2023); the comparison of Sec. 8.5 therefore isolates the effect of the input representation from the choice of generative engine.

No likelihood evaluations, simulator gradients, or per-dataset optimization are required at any stage; the simulator is executed only forward, during training-set generation.

What is and is not guaranteed. The construction has an exact consistency property.

Proposition 1 (consistency in the idealized limit)**.** *Let $(m, \mathcal{D}) \sim p(m)\,p(\mathcal{D} \mid m)$ and let z_1 be given by* (2)*. If the loss* (3) *is minimized over all measurable velocity fields, then for $p(\mathcal{D})$-almost every $\mathcal{D}$ the flow of the minimizer transports $\mathcal{N}(0, I)$ at $t = 0$ to the conditional law $p(z_1 \mid \mathcal{D})$ at $t = 1$; equivalently, the sampling procedure above draws exactly from the posterior (Lipman et al., 2023; Wildberger et al., 2023).*

Two things are not guaranteed, and they shape the rest of the paper. First, with a finite network, finite simulation budget, and a numerical ODE solver, no calibration guarantee survives; the residual error is an empirical quantity, which is why Sec. 6 builds instruments to measure it rather than assuming it away. Second, the expectation in (3) weights approximation error by the probability of the data: directions of parameter space in which the posterior is nearly flat contribute little to the loss and are learned last. Both effects are visible and quantified in the experiments — the first as width ratios slightly above one that decrease with budget, the second as the along-ridge bias of the Fisher–KPP posterior (Sec. 8.3).

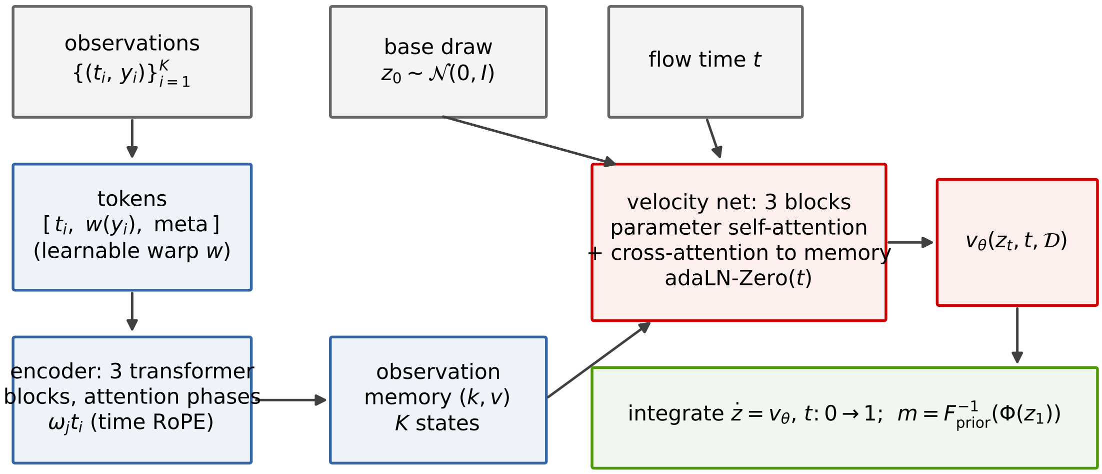

Figure 1: Architecture. Left: each observation (t_i, y_i) becomes a token $[\,t_i,\ w(y_i),$ metadata$\,]$ with a learnable monotone transform w (Sec. 4.3); a three-block transformer encoder, whose attention phases are computed from the physical times t_i (Sec. 4.2), produces a memory of K states. Right: the d parameter components of the flow state z_t form d tokens processed by three velocity blocks, each applying self-attention across parameter slots and cross-attention into the observation memory, modulated by the flow time through zero-initialized adaptive layer norms. Integrating the resulting velocity field from a Gaussian draw yields a posterior sample.

4 Conditioning architecture

The velocity field $v_\theta(z, t, \mathcal{D})$ must be conditioned on a dataset that is a *set of time-stamped measurements*, and this drives the architecture. Four requirements follow directly from the problem statement: the network must accept any number K of observations; its output must not depend on the order in which observations are stored, since the times t_i carry all ordering information; the times are irregular real numbers, not grid indices; and, because posterior width is governed by K, the dependence on K must be represented accurately, not merely tolerated. The first three requirements rule out the fixed-length-vector conditioners that are the default in NPE implementations; they are naturally met by a transformer operating on one feature vector per observation. Throughout this section the term *token* means exactly that: the feature vector representing one observation.

4.1 Overall scheme

Figure 1 shows the two halves of the model.

Encoder. Observation i becomes the token $x_i = [\,t_i,\ w(y_i),$ sensor id, design metadata$\,]W_{\text{emb}}$, where w is the learnable monotone transform of Sec. 4.3. Three standard transformer blocks (multi-head self-attention followed by a two-layer MLP, pre-norm residual form) exchange information among the K tokens; attention weights are $\text{softmax}_j(q_i^\top k_j/\sqrt{d_h})$ with queries and keys rotated by time-dependent phases as defined in Sec. 4.2. The result is a memory of K encoded states.

Velocity network. The d components of the flow state z_t are treated as d tokens. Each of three velocity blocks applies self-attention across the parameter slots — this is where posterior correlations between parameters are expressed — and cross-attention from the parameter slots into the K-state observation memory, which is how information about the data, at any K, reaches the flow. The flow time t enters through adaptive layer normalization: each sub-layer computes $h \mapsto h + g \odot f\big((1+\gamma) \odot \text{LN}(h) + \beta\big)$ with (γ, β, g) produced from a sinusoidal embedding of t by a zero-initialized linear map (Peebles and Xie, 2023), so the untrained network starts as an identity-like map and departs only as driven by the loss. The default model uses width 64, four heads, three encoder and three velocity blocks: 4.8×10^5 parameters in total. Section 8.3 shows that doubling width and depth changes no benchmark result, so the default is deliberately small.

This scheme was not the first attempt but the survivor of a sequence of measured alternatives; the failed ones are retained in the package as reproducible controls and appear as rows of Table 3. The remainder of the section describes the three components that make the scheme work at benchmark accuracy: the time encoding, the input transform, and the design-size features — plus one optional token construction that accelerates a special but common case.

4.2 Attention phases from observation times

Transformers are made order-aware by positional encodings. The rotary scheme (Su et al., 2024) splits each head dimension into two-dimensional subspaces and rotates queries and keys in subspace j by an angle proportional to the token's *position* n in the sequence: $q_i \mapsto R(n_i\omega_j)\, q_i$, $k_i \mapsto R(n_i\omega_j)\, k_i$, with a geometric frequency bank ω_j, so that $q_i^\top k_l$ depends only on $n_i - n_l$. For irregularly sampled data the storage position is meaningless, and we replace it by the physical observation time:

$$q_i \;\mapsto\; R(\omega_j t_i)\, q_i, \qquad k_i \;\mapsto\; R(\omega_j t_i)\, k_i, \tag{4}$$

so attention logits depend on the time differences $t_i - t_l$ and on nothing else about storage order. The encoder is then permutation-invariant by construction (verified to 10^{-8} under random token permutations), and design amortization involves no positional bookkeeping at all.

The measurement that forced this choice is worth recording. An earlier configuration used storage-index rotations, trained with long token sequences subsampled in place (kept tokens retained their spread-out indices), and evaluated on contiguously packed short sequences. Every component was individually correct, yet the mismatch between the positional distributions seen at training and evaluation produced systematic posterior biases of +1.1 to +3.3 standard deviations (first row of Table 3). Index-based encodings create an implicit train/test contract on token placement; time-based phases eliminate the contract rather than manage it.

4.3 Learnable coordinate transform

The values y_i enter tokens through a monotone map

$$w(x) \;=\; \sum_{k=1}^{8} a_k \;\mathrm{sign}(x)\log\!\big(1 + |x|/s_k\big), \qquad a_k \geq 0, \tag{5}$$

a positive mixture over a geometric bank of scales s_k spanning eight octaves; the same construction (with its own weights) is applied to time gaps where they appear. Since $\log(1 + |x|/s) \approx |x|/s$ for $|x| \ll s$ and $\approx \log|x| - \log s$ for $|x| \gg s$, the mixture interpolates continuously between the identity and the logarithm, and the network can learn, per problem, the coordinate in which the observed process is well-conditioned — for geometric Brownian motion it learns the log-price coordinate in which increments become homoscedastic.

A natural first candidate for this role is the Yeo–Johnson transform (Yeo and Johnson, 2000), the standard one-parameter family of monotone power transforms used in applied statistics to normalize skewed data; it contains the identity ($\lambda = 1$) and, for positive data, the logarithm ($\lambda \to 0$) in its parameter space. Used as the trainable input transform on GBM, it failed: across seeds, gradient descent left λ near its initialization, and the volatility bias remained at +0.48 posterior sd — the same value as with no transform. The mixture (5) reaches the log coordinate reliably because all of its components produce gradients at initialization, whereas moving λ requires traversing a region of the loss surface with no useful signal. We record this ablation because it illustrates the design rule used throughout the package: what matters is not whether the correct transformation exists in the hypothesis class, but whether gradient descent reaches it from the initialization.

4.4 Optional increment tokens for directly observed diffusions

For scalar SDEs whose state is observed directly — no latent components — the package offers a second token construction: after sorting by time, consecutive observations contribute

$$\big[\, t_{(i)},\; w(y_{(i)}),\; \Delta w_i,\; (\Delta w_i)^2,\; w_t(\Delta t_{(i)}) \,\big], \qquad \Delta w_i = w(y_{(i+1)}) - w(y_{(i)}).$$

The motivation is elementary: for a Markov observed process the likelihood factorizes over consecutive pairs, transition densities of diffusions depend on increments and gaps, and squared increments carry the quadratic-variation information that identifies diffusion coefficients. This is an inductive bias of the same character as convolutions for images — useful where its assumption holds, not a requirement of the method: the point-token path of Sec. 4.1 is fully general, and Table 3 quantifies what the increment tokens buy at fixed budget (the dense-design width drops from 1.49 to 1.27 before the training procedure closes the rest).

The rule for choosing between the constructions — increment tokens exactly when the observed process is Markov — was tested in both directions: an increment-token model class transfers across Markov systems without adjustment (GBM to OU), and on the Heston model, whose observed price is not Markov because volatility is latent, increment and point tokens perform identically across all 25 evaluation cells, as the factorization argument predicts. The package selects the construction from a per-problem flag.

Two additions make increment tokens robust on heavy-tailed processes (the measurements are in Sec. 9). Squared increments are logarithmically compressed before feature normalization. And the normalization is made dataset-adaptive by feature-wise affine modulation (FiLM; Perez et al., 2018): pooled robust statistics s of the token set (computed over non-padding positions only) are mapped by a zero-initialized MLP to per-channel scales and shifts,

$$h_c \;\mapsto\; \big(1+\gamma_c(s)\big)\, h_c + \beta_c(s), \tag{6}$$

applied after the running feature normalization. Zero initialization keeps the working baseline at the start of training; the ordering (normalize, then modulate) matters and is discussed with the failure it prevents in Sec. 9.

4.5 Design-size features

Softmax attention computes normalized weighted means, which are nearly invariant to the number of tokens — yet posterior width is governed by K. The deficit is measurable by linear probes from the pooled encoder output: all architecture variants of Table 3 recover posterior-mean statistics ($R^2 = 0.83$–0.94), while $R^2(\log K)$ ranges from 0.985 to 0.498 and ranks the variants exactly in the order of their width errors. The repair is to supply the count: an explicit $\log K$ token feature, or two learned register tokens (Darcet et al., 2024) that accumulate it. On the hardest cell ($K = 100$) the explicit feature reduces the width ratio from 1.95 to 1.49 with bias unchanged; the residual is closed by the training procedure of the next section.

5 Training procedures

Common settings. Adam with cosine decay and linear warmup, base learning rate 3×10^{-4}, batch 64, gradient clipping, EMA decay 0.999. Each optimizer step draws fresh flow times t (stratified) and fresh base noise. Fixed-design models train on n simulated datasets for a fixed number of epochs (the gallery of Sec. 8.1 uses $n = 40{,}000$ and 35 epochs).

Design amortization. Training data are simulated once on a fine grid; observation designs are then resampled *from the stored fine trajectories at every optimizer step.* The procedure has three components, each fixed by a controlled comparison on three systems (GBM, Ornstein–Uhlenbeck, FitzHugh–Nagumo):

1. *Fresh designs at every step.* Re-drawing each trajectory's observation subset at each step makes design memorization impossible and lets one simulation serve arbitrarily many designs. The alternative — one fixed design per trajectory — closes the dense-design error at long budgets but overfits the design set, producing overconfidence at sparse K (width ratio 0.86 against the exact reference).

2. *The design-size law.* K is drawn 50% log-uniform on $[K_{\min}, K_{\max}]$ and 50% uniform on the dense half of the range. Log-uniform sampling alone places $\approx 6\%$ of the mass at $K \geq 100$ and starves the dense regime, which appears as inflated widths precisely there; the mixture alone halves the dense-design excess (GBM 1.27 $\to$ 1.14, OU 1.34 $\to$ 1.14; the FitzHugh–Nagumo dense-K SBC failure at $p < 10^{-3}$ disappears, $p = 0.54$).

3. *Budget scaling.* When higher accuracy is required, simulations and optimizer steps are scaled together at constant epochs; Sec. 8.3 shows the residuals then follow power laws in this single budget variable.

With this procedure, the GBM posterior width matches the exact per-design posterior to 5–7% uniformly over $K = 5, 9, 20, 100$ (Fig. 3).

6 Evaluation methodology

Two instruments with different roles are used throughout, plus a diagnostic probe.

6.1 Simulation-based calibration as a screen

SBC (Talts et al., 2018) tests a necessary condition: for $m^* \sim p(m)$ and $\mathcal{D} \sim p(\mathcal{D} \mid m^*)$, the rank of each component of m^* among posterior samples must be uniform. We run SBC per parameter (uniformity test on 500 simulated datasets with 200 posterior draws each unless stated; $p > 0.05$ passes) and report coverage of central 50% and 90% intervals.

The power of the test at this size is limited, and because our conclusions depend on knowing that limit, we measured it: on cells where an exact reference is available, SBC at 300–500 datasets fails to reject posterior width ratios up to ≈ 1.3, and its per-cell p-values are unstable between retrainings when residual biases are in the range 0.1–0.2 posterior standard deviations. SBC also cannot penalize a posterior that ignores the data and returns the prior; the prior-mean and ridge controls of Sec. 7 exist for that reason. SBC is therefore used as a cheap screen, never as a verdict.

6.2 Reference posteriors as the record

Wherever a tractable likelihood exists, we compare distributions on the same observed points using two statistics per parameter: the signed error of the posterior mean in units of the reference posterior standard deviation (*bias*) and the ratio of standard deviations (*width*; 1 is exact, > 1 conservative, < 1 overconfident). References are of three kinds (full details in Appendix B):

- *Closed form.* The linear-Gaussian benchmark (exact Gaussian posterior) and geometric Brownian motion, whose log-returns give a conjugate normal–inverse-χ^2 posterior importance-corrected to the box prior.
- *Exact-likelihood MCMC.* An adaptive random-walk Metropolis sampler (after Haario et al., 2001): proposal scale adapted to a 0.25 acceptance target during burn-in, proposal shape replaced by the empirical covariance Cholesky factor rescaled to $2.38/\sqrt{d}$, adaptation frozen before the retained phase, out-of-box proposals rejected (which samples the box-truncated target exactly). Likelihoods are exact per-gap transition densities: Gaussian Euler–Maruyama chain transitions for OU (matching the simulator that generated the data, plus the stationary density of the informative initial state), a Poisson-mixture transition density for the Merton jump-diffusion, log-normal residuals around the Bateman curve for pharmacokinetics, and a Gaussian observation model around the deterministic PDE solve for Fisher–KPP — in the last case one likelihood evaluation is one PDE solve.
- *Chain validation.* A reference is used only after validation. Every MCMC reference in this report is run at least twice from well-separated starting points, and the between-chain discrepancy of posterior means (in posterior-sd units) is reported with the result it certifies. Where a closed form and a chain both exist, the chains reproduce it to a worst mean discrepancy of 0.099 sd and width ratios 0.91–1.03 over 24 comparisons.

Twice during development the screen and the record contradicted each other; both times the contradiction exposed an implementation error (a padding mask leaking into set-level statistics; the positional contract of Sec. 4.2). We consider the redundancy a feature of the methodology rather than an inefficiency.

6.3 Encoder probing

To separate encoder failures from flow failures, we fit ridge regressions from the pooled encoder output to known sufficient statistics of the problem. High probe R^2 with poor posterior accuracy localizes the defect downstream of extraction. In the Merton analysis of Sec. 9, the failing configuration extracted robust volatility statistics *better* than the healthy one ($R^2 = 0.967$ on truncated realized variance), redirecting the repair from the encoder's capacity to the stability of its input representation.

system	type	d	obs.	design	reference posterior
linear-Gaussian	linear map	4	full	fixed	exact Gaussian
Ornstein–Uhlenbeck	SDE	2	full	fixed	exact-likelihood MCMC
geometric Brownian motion	SDE	2	full	fixed	conjugate closed form
Cox–Ingersoll–Ross	SDE	3	full	fixed	— (Euler pseudo-MLE)
double well	SDE	3	full	fixed	—
stochastic Lotka–Volterra	2-d SDE	4	full	fixed	—
FitzHugh–Nagumo	2-d SDE	4	v only	fixed	—
SEIR	4-d SDE	5	I, D	fixed	—
SINDy-SDE (cubic drift)	SDE	5	full	fixed	—
Heston	2-d SDE	5	price only	variable	—
Merton jump-diffusion	jump SDE	5	full	variable	Poisson-mixture MCMC
Hénon–Heiles	Hamiltonian ODE	3	q_1 only	variable	—
Hodgkin–Huxley	4-d ODE	4	V only	variable	—
pharmacokinetics (Bateman)	ODE (closed form)	3	concentration	variable	log-normal MCMC
Fisher–KPP	PDE (64-point grid)	2	3 sensors	variable	PDE-likelihood MCMC

Table 1: The benchmark suite. "obs." lists the observed coordinate when the state is only partially visible; "design" distinguishes the fixed-grid gallery from variable-design systems, where every dataset carries its own set of (time, sensor) observation points. Dashes in the last column indicate systems where no tractable reference exists — these are evaluated by SBC and controls only, which is precisely the regime that motivates carrying exact-reference systems in the same suite.

7 Benchmark suite

The suite contains fifteen systems (Table 1): nine with a fixed observation design used for the calibration gallery, and six with variable designs used for design amortization. Every system ships with a classical estimator (`sota()`) and, where a tractable likelihood exists, a reference posterior. Full equations, priors, and observation models are given in Appendix A.

Controls and classical baselines. Every result is read against: (i) a prior-mean predictor (exactly 25% mean absolute error under a uniform prior) and a degree-2 ridge regression on ∼20 generic summary statistics — these two detect prior-returning posteriors that SBC passes; (ii) the per-system classical estimator: exact MLE for OU and GBM, Euler pseudo-likelihood for CIR, Kramers–Moyal and SINDy-style regression for drift discovery (Brunton et al., 2016), multi-start nonlinear least squares for FitzHugh–Nagumo, SEIR, and Lotka–Volterra, and the exact posterior mean for the linear-Gaussian case; (iii) the reference posteriors of Sec. 6.2.

8 Experimental results

8.1 Fixed-design gallery

Table 2 summarizes the nine fixed-design systems. One model configuration (the default of Sec. 4) is trained per system on 40,000 simulations for 35 epochs (about 11 minutes per system on one H200 GPU); SBC uses 500 datasets with 200 posterior draws. Of 32 parameters, 31 pass; mean central-interval coverage error is 1.4 percentage points. The failing cell in this run is the mean-level parameter b of Cox–Ingersoll–Ross; in an independent earlier training run of the same configuration on a CPU, 31 of 32 parameters passed as well, but the failing cell was the β parameter of stochastic Lotka–Volterra at $p = 0.04$ while Cox–Ingersoll–Ross passed at $p \geq 0.33$. The identity of the marginal cell is not stable across retrainings — which is exactly the behavior predicted by the power analysis of Sec. 6 for residual defects of 0.1–0.2 posterior sd, and one reason SBC alone is not used as a verdict. On the systems with exact references, the learned posteriors carry 89–90% of the available information (width relative to the exact posterior). Per-parameter p-values for the run shown are listed in Appendix A. Figure 2 shows the SBC rank distributions themselves — empirical CDFs of the normalized ranks for all 32 parameters, with the pointwise band that a uniform distribution would occupy at this sample size — so the reader can judge the calibration claim directly rather than through p-values alone.

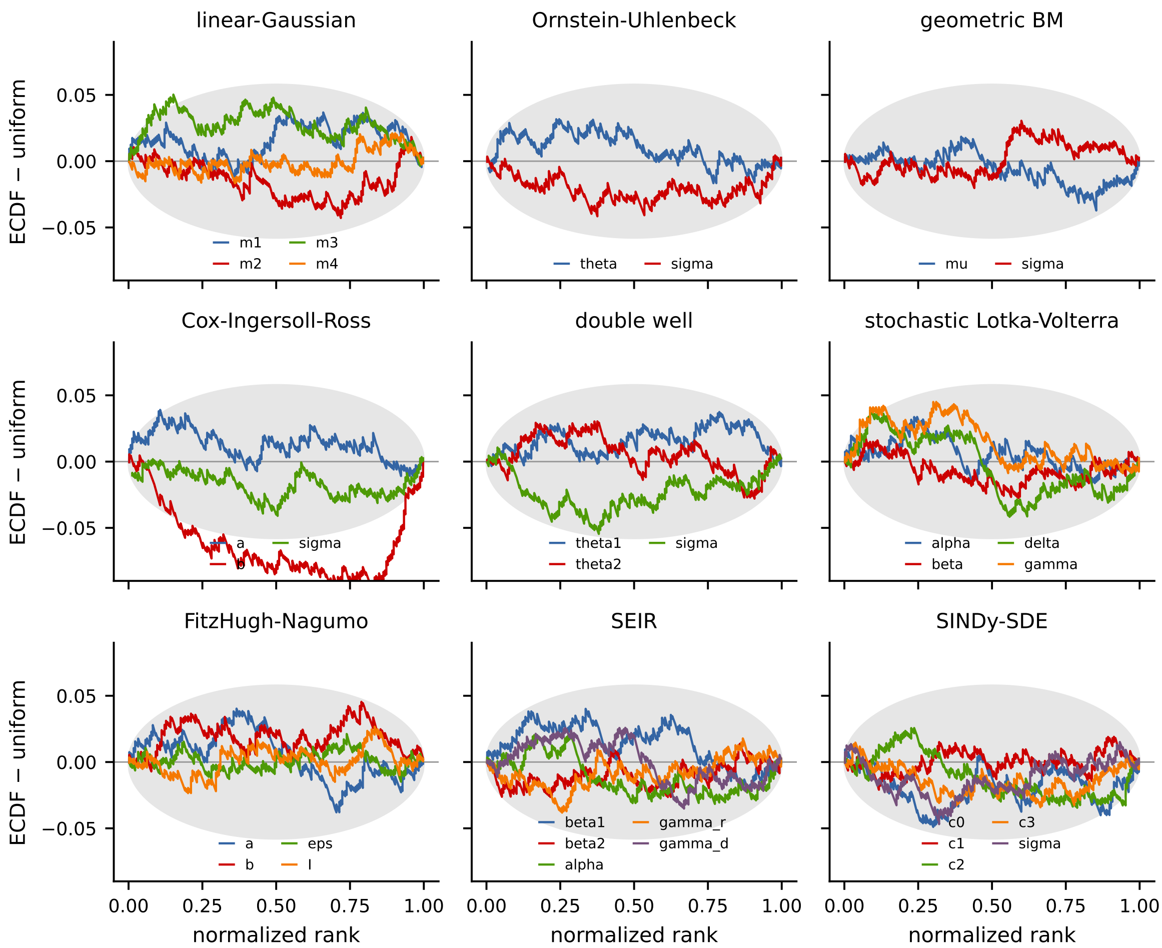

Figure 2: Simulation-based calibration for the fixed-design gallery: empirical CDFs of normalized SBC ranks (500 datasets, 200 posterior draws) for every parameter of every system, shown as deviation from the uniform diagonal, with a 99% pointwise band. Curves inside the band are consistent with calibration; the single failure of this run (Cox–Ingersoll–Ross b) is visible as the curve leaving the band.

8.2 Variable designs

Architecture comparison under a common protocol. Table 3 and Fig. 3 compare encoder variants for design amortization on GBM, trained at $K \sim \text{log-uniform}[2, 128]$ random-time designs and evaluated against the exact per-design posterior (96 datasets per bucket). Three observations organize the table. First, index-based positions combined with in-place subsampling during training (leaving kept tokens at spread positions while inference packs contiguously) produce biases of $+1.1$ to $+3.3$ posterior sd with no individually incorrect component — the positional pathology of Sec. 4.2; repacking or replacing positions with time phases removes the bias entirely. Second, all point-token variants share inflated widths at dense designs, in proportion to how poorly their pooled representation encodes $\log K$ (rightmost column; the probe of Sec. 4.5) — including two attention mechanisms specifically constructed to exploit continuous time (a learned per-head time-kernel logit bias in the spirit of relative-position methods, and edge-message attention carrying gap-modulated increments on all pairs). Third, pair tokens with time-phase attention, plus the training procedure of Sec. 5, bring every bucket to within 7% of the exact width.

Variable-design zoo. With the class rule and training procedure fixed, six further systems were trained without per-system adjustments. Heston (five parameters, latent volatility): all 25 SBC cells pass, and pair tokens give no advantage over point tokens, consistent with the class boundary (Sec. 4.4). Merton jump-diffusion: 25 of 25 cells after the heavy-tail treatment of Sec. 9, including dense-design cells that failed at $p < 10^{-3}$ before it; the residual width sits on the budget curve below. Hénon–Heiles (parameters of the potential of Lubich et al. 2015, observing one coordinate at random times): calibrated at every design size down to $K = 6$, where the posterior widens honestly rather than degrading. Hodgkin–

system	d	coverage err.	SBC pass	cov. 50%	cov. 90%
linear-Gaussian	4	1.3 pp	4/4	51%	90%
Ornstein–Uhlenbeck	2	1.8 pp	2/2	47%	88%
SEIR	5	1.5 pp	5/5	49%	90%
geometric Brownian motion	2	1.5 pp	2/2	50%	90%
Cox–Ingersoll–Ross	3	1.5 pp	2/3	49%	88%
double well	3	0.8 pp	3/3	50%	90%
stochastic Lotka–Volterra	4	1.9 pp	4/4	48%	88%
FitzHugh–Nagumo	4	1.1 pp	4/4	51%	90%
SINDy-SDE (cubic drift)	5	1.4 pp	5/5	50%	89%
total / mean	32	1.4 pp	31/32	49%	89%

Table 2: Fixed-design gallery: one default model per system, 40,000 simulations, 35 epochs. Coverage error is the mean absolute deviation of central-interval coverage from its nominal level; "SBC pass" counts parameters with uniformity $p > 0.05$ (500 datasets $\times$ 200 draws).

	σ width ratio at K				
encoder variant	5	9	20	100	$R^2(\log K)$
point tokens, index positions, spread subsampling	0.94*	1.13*	1.42*	2.08*	—
point tokens, index positions, contiguous	1.05	1.17	1.13	1.73	0.957
point tokens, time-kernel attention bias	1.03	1.14	1.31	2.51	0.712
point tokens, edge-message attention	1.03	1.15	1.29	2.45	0.498
point tokens, time phases	1.01	1.10	1.14	1.95	—
+ explicit $\log K$ feature	1.01	1.05	1.08	1.49	—
pair tokens, time phases	1.02	1.02	1.02	1.27	0.985
pair tokens + full training procedure	1.06	1.06	1.05	1.07	—

Table 3: GBM with variable designs: posterior width for σ against the exact per-design posterior (96 datasets per bucket) and, where measured, the probe R^2 for $\log K$ from the pooled encoder output. *This row also carries biases of +1.09 to +3.30 posterior sd (the positional train/test mismatch of Sec. 4.2); all other rows have biases consistent with zero.

Huxley (Hodgkin and Huxley, 1952): calibrated on mixed designs; one dense-design cell shows the budget signature. Pharmacokinetics: at $K = 6$ irregular blood draws, bias -0.01 to -0.06 sd and widths 1.09–1.33 against exact-likelihood MCMC. Fisher–KPP: the posterior concentrates on the ridge $D \cdot r \approx \text{const}$; the residual along-ridge bias decays with budget (next subsection).

8.3 Residuals scale as power laws in the training budget

Accuracy against the exact references is a smooth, monotone function of the training budget, and we report it systematically rather than at a single operating point. Figure 4 shows posterior width and bias against the reference for three systems (GBM, OU, Fisher–KPP with the fixed design of Sec. 8.5) over a grid of budgets from 5,000 to 40,000 simulations with optimizer steps scaled proportionally. Posterior widths converge to one from above on all three systems (Fisher–KPP: 1.45 $\to$ 1.01). Posterior-mean biases behave differently by identifiability: on GBM and OU they stay within ± 0.1 sd throughout, while on Fisher–KPP the bias lies along the weakly identified ridge direction and does not decay monotonically — independently trained models land at different points within ± 0.5 sd along the ridge, consistent with the flat direction carrying the weakest training signal (Sec. 3). Under the variable-design training procedure, which averages every optimizer step over fresh observation designs, the same ridge bias does decay with budget. Beyond it, the two residual defects of the variable-design suite — inflated width on the hardest Merton cell and along-ridge bias on Fisher–KPP under random designs — respond to the same budget lever (Fig. 5). The Merton width excess follows $\propto 1/\text{budget}$ (0.246 $\to$ 0.120 $\to$ 0.059 at $\times 1, \times 3, \times 6$); the KPP ridge bias follows $\approx \text{budget}^{-1/2}$ (+0.41 $\to$ +0.23 $\to$ +0.14 sd along the ridge, with the sharply identified front-speed component decaying in parallel). Two negative controls exclude the obvious alternatives: doubling network width and depth at fixed budget changes the Merton cell by less than 0.02 in width ratio (1.103 vs 1.120 at $\times 3$), and refining the sampling ODE solver from 20 midpoint steps to 60 RK4 steps changes nothing measurable. The weakly identified posterior direction carries the weakest regression signal in (3) and converges last; we find no evidence of a floor within the measured range.

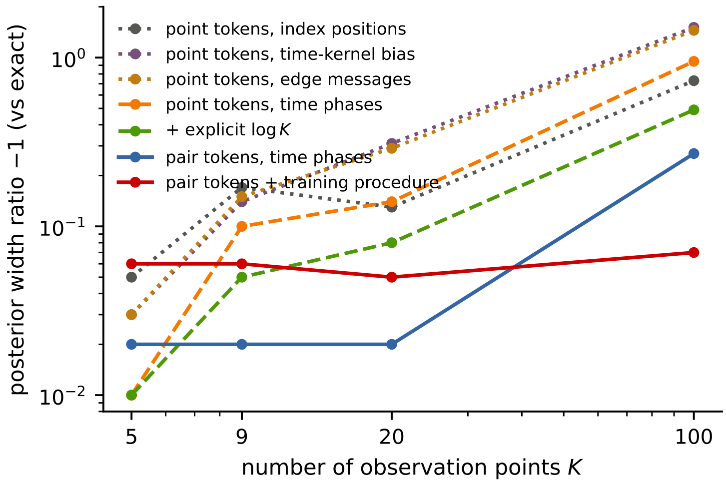

Figure 3: Posterior width for the GBM volatility against the exact per-design reference, as a function of design size K, for the encoder variants of Table 3. Dotted: point-token variants without set-size information; dashed: partial repairs; solid: pair tokens, and pair tokens with the full training procedure of Sec. 5.

8.4 Posterior geometry against validated references

Scalar bias and width statistics summarize accuracy but hide the joint structure of the posterior. Figure 6 overlays two-dimensional posterior samples of the trained models on their validated references for three systems where the reference is nontrivial: the Merton jump-diffusion (volatility–jump-rate plane, where the two parameters compete to explain variance), pharmacokinetics at $K = 6$ irregular blood draws (absorption–elimination plane, the data-poor clinical regime), and Fisher–KPP (diffusion–reaction plane, where the posterior concentrates on the ridge $Dr \approx$ const set by the front speed). The learned posteriors reproduce the location, scale, orientation, and curvature of the references, including the ridge geometry; the corresponding scalar statistics appear in the tables of this section and in Fig. 4.

8.5 Comparison with existing implementations

Protocol. We compare against the directly comparable amortized estimators available in maintained packages: NPE with its default masked autoregressive flow (Papamakarios et al., 2017) and FMPE (Wildberger et al., 2023) from `sbi` 0.27 (Boelts et al., 2024); the coupling-flow and flow-matching networks of BayesFlow 2.0 (Radev et al., 2020); and, because it is the closest architecture to ours in the literature, a port of the Simformer (Gloeckler et al., 2024) built from its authors' minimal example (transformer–diffusion over the joint, random condition masks, their architecture and SDE settings, full 75,000-step training). All arms receive the same simulation budget (20,000 draws from the same prior and simulator), run with their package defaults and their own stopping rules, and are scored on the same test datasets against the same reference posteriors, using the bias and width statistics of Sec. 6.2. Because these estimators consume fixed-length vectors, the comparison uses fixed observation designs. Three systems are used, chosen to separate the possible explanations of any performance gap: a multiplicative process where input conditioning is hostile (GBM, exact conjugate reference), an additive process where it is comparatively benign (OU, exact-likelihood MCMC reference, two validated chains per dataset), and a PDE where every likelihood evaluation requires a forward solve (Fisher–KPP), so that the cost relationship between reference and amortized inference is inverted. Where informative, external arms are additionally given hand-derived inputs — log-prices, or per-gap log-returns, the exact coordinates of the GBM likelihood factorization — to separate the effect of the input representation from the generative engine.

Multiplicative process (GBM). Table 4 and Fig. 7 give the results; timing columns are wall-clock on one laptop CPU (Apple M-series; Simformer rows on an H200 GPU, so their timings are not comparable with the rest while their accuracy rows are). Three regularities are visible. First, with raw prices as input, every external configuration inflates the volatility posterior — widths 1.85 (BayesFlow coupling) to 3.16 (Simformer port) — irrespective of the generative engine; `sbi` itself emits a warning about precision loss during z-scoring on this data, which is a fair description of the mechanism (Sec. 9). Second, supplying

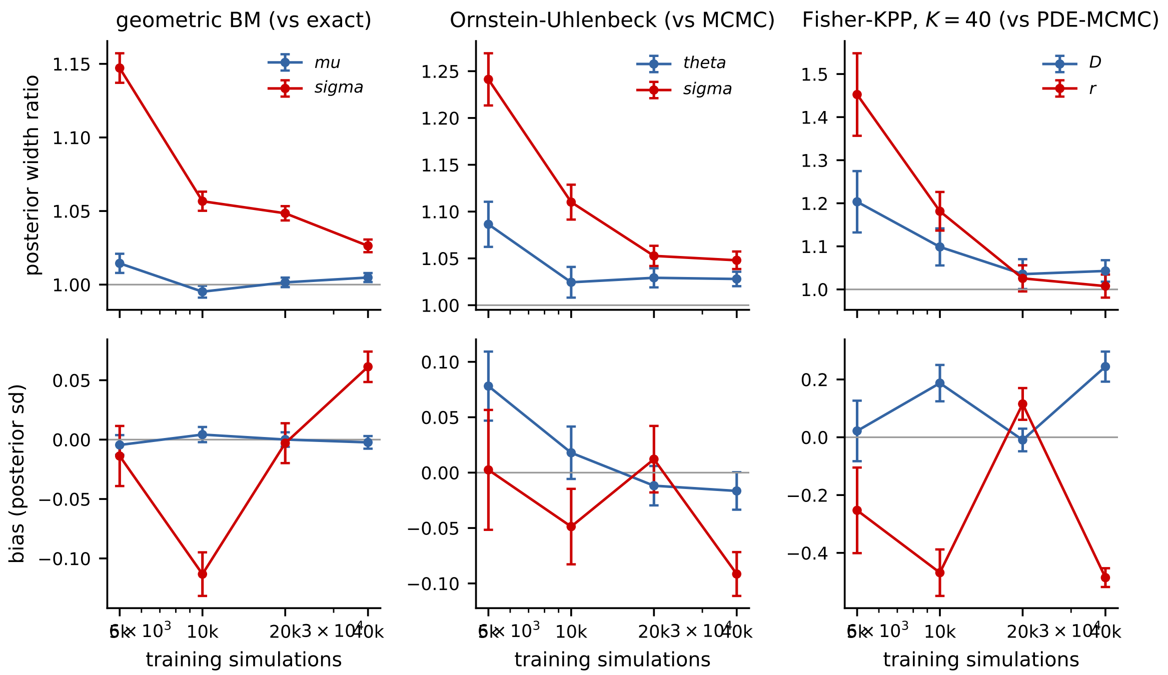

Figure 4: Convergence to the reference posterior with training budget on three systems (columns): posterior width ratio (top) and posterior-mean bias (bottom) per parameter, over budgets of 5,000–40,000 simulations with optimizer steps scaled proportionally. References: exact conjugate posterior (GBM), exact-likelihood MCMC (OU), PDE-likelihood MCMC (Fisher–KPP, fixed $K = 40$ design).

hand-derived coordinates does not produce convergent behavior: with identical log-return inputs, NPE remains 44% over-dispersed, FMPE is nearly exact (1.09), both BayesFlow arms become overconfident (widths 0.42–0.85), and the Simformer port acquires a −0.48 sd bias. Each of these defects is invisible without an exact reference, and each lies within or near the blind region of SBC at practical sizes (Sec. 6). Third, at fixed input coordinates the flow-matching engine outperforms the autoregressive flow (1.09 vs 1.44), consistent with the engine choice of Sec. 3, but no external arm matches the learnable representation operating on raw data (1.05). The verdicts are stable across devices and seeds: re-running the sbi and BayesFlow arms on an H200 GPU with fresh seeds reproduces every failure direction (NPE 2.48, FMPE 2.30, BayesFlow coupling 1.80 on raw input; BayesFlow overconfident at 0.40–0.72 on log-returns).

Additive control (Ornstein–Uhlenbeck). The same protocol on OU — an additive process with no floating scale — separates the candidate explanations of the GBM results (Table 5). The reference is exact-likelihood MCMC (per-gap Euler–Maruyama transitions plus the stationary initial density; 62 ms per chain; two-chain validation: worst mean discrepancy 0.29 sd, mean 0.08 sd over 100 datasets). The external estimators improve markedly relative to GBM: the volatility bias is no longer significant ($+0.19 \pm 0.19$ for NPE, $+0.08 \pm 0.12$ for FMPE), and the width inflation drops from 2.5–3.2 to 1.9 (NPE) and 1.6 (FMPE). This is the pattern the conditioning analysis predicts: the form-1 pathology of Sec. 9 is absent by construction, and what remains is the general difficulty of assembling quadratic-variation statistics from a raw high-dimensional vector — present but smaller. `amortix` reads 1.03/1.05 on the same data.

Simulator-limited regime (Fisher–KPP). The third system inverts the cost structure of the first two: every likelihood evaluation requires solving the forward model. A one-dimensional reaction–diffusion equation is inexpensive in absolute terms (16 ms per solve here); what makes the case informative is the *ratio* between reference and amortized inference, because both sides scale linearly with the solve cost — a chain of n likelihood evaluations costs n solves per dataset whatever the forward model, while the amortized cost per dataset does not involve the simulator at all. The measured ratio below transfers to genuinely expensive simulators, where the same reference sampler would take days per dataset. A single fixed design of $K = 40$ random (time, sensor) points is shared by all datasets and methods; the reference is the PDE-likelihood sampler of Sec. 6.2 with two validated chains per dataset (271 s per dataset for the pair; two-chain validation over 32 datasets: worst mean discrepancy 0.22 sd, mean 0.07 sd). Table 6 gives

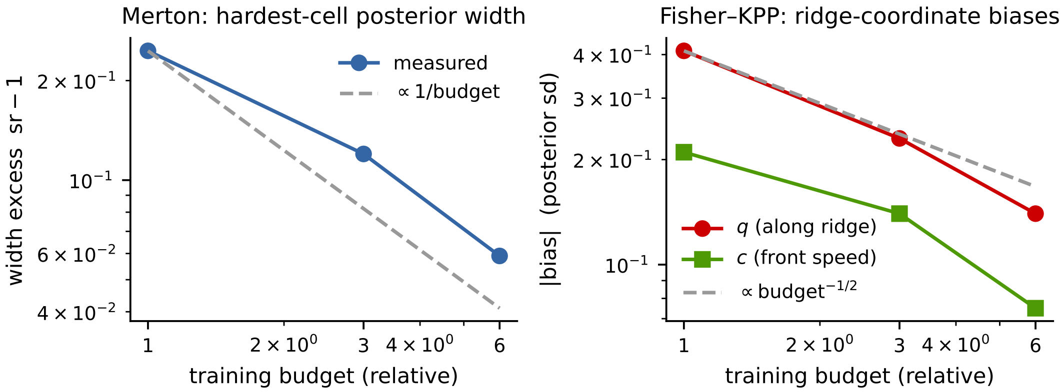

Figure 5: The two residual defects of the suite against training budget (simulations and steps scaled together; model size fixed). Left: Merton width excess on the hardest evaluation cell, with the 1/budget guide. Right: Fisher–KPP posterior-mean biases in the ridge coordinate system, with the budget$^{-1/2}$ guide.

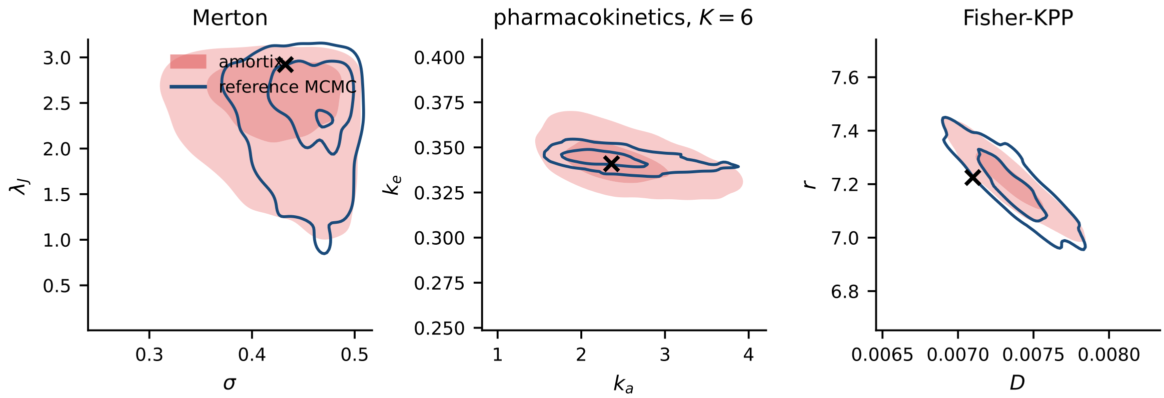

Figure 6: Joint posteriors of `amortix` (filled contours) against validated reference posteriors (open contours; exact-likelihood MCMC, two chains per dataset) on one test dataset per system. Left: Merton jump-diffusion (σ, λ_J). Middle: pharmacokinetics (k_a, k_e) at $K = 6$ observations. Right: Fisher–KPP (D, r), whose posterior concentrates on the front-speed ridge. Crosses mark the true parameters.

the results. `amortix` matches the reference (1.02/1.04 width, biases within 0.13 sd) at 31 ms per dataset; its entire training cost (379 s) equals the reference cost of fewer than three datasets, so amortization pays for itself almost immediately in any repeated-inference setting. NPE at defaults shows moderate width inflation (1.26–1.28); FMPE at defaults fails outright on the diffusivity ($+7.3 \pm 2.7$ sd bias at 12.5× width) while remaining reasonable on the reaction rate — a further instance of default configurations behaving discontinuously across parameters, and detectable here only because the expensive reference was computed.

Scope of the comparison. Every external arm ran at its package defaults under our simulation budget; specialists could improve each with problem-specific embedding networks or summary statistics. The comparison is designed to measure exactly that dependence: what accuracy is delivered when the input representation is not hand-engineered per problem. The Simformer arm in particular is a port of a minimal example rather than a tuned configuration, and its numbers should be read as representative of that recipe, not of the method's ceiling. Conversely, the variable-design capability of Sec. 8.2 is not exercised here at all, since the external estimators consume fixed-length inputs by default.

8.6 Computational cost

The default model trains in minutes to tens of minutes per system: on one H200 GPU, a variable-design zoo system at base budget (20,000 simulations, 12,000 steps) takes 8–12 minutes, and the ×6 budget behind the sharpest numbers of Fig. 5 takes about 50 minutes; on a laptop CPU the fixed-design gallery

method (input)	train	inference	μ: bias / width	σ: bias / width
`amortix` CFM, raw prices	488 s	426 ms	+0.00 / 1.00	−0.00 / 1.05
`sbi` NPE, raw prices	29 s	13 ms	−0.02 / 1.06	$+0.45 \pm 0.19$ / 2.64
`sbi` NPE, log-prices	34 s	12 ms	−0.04 / 1.01	$+0.32 \pm 0.17$ / 2.29
`sbi` NPE, log-returns	10 s	13 ms	+0.07 / 1.07	$+0.13 \pm 0.10$ / 1.44
`sbi` FMPE, raw prices	27 s	370 ms	−0.03 / 1.03	$+0.20 \pm 0.16$ / 2.46
`sbi` FMPE, log-returns	17 s	333 ms	+0.01 / 1.03	-0.02 ± 0.07 / 1.09
BayesFlow 2 coupling, raw prices	384 s	4 ms	−0.04 / 0.95	-0.13 ± 0.11 / 1.85
BayesFlow 2 coupling, log-returns	459 s	6 ms	−0.01 / 0.42	-0.15 ± 0.05 / 0.54
BayesFlow 2 flow matching, raw prices	220 s	2220 ms	+0.06 / 1.00	-0.36 ± 0.11 / 1.99
BayesFlow 2 flow matching, log-returns	296 s	2311 ms	−0.04 / 0.46	-0.30 ± 0.06 / 0.85
Simformer port, raw prices	591 s	1057 ms	−0.04 / 1.08	-0.30 ± 0.20 / 3.16
Simformer port, log-returns	548 s	$\sim 10^3$ ms	−0.08 / 1.10	-0.48 ± 0.02 / 1.36
exact-likelihood MCMC	—	42 ms	reference (chain check: −0.04 / +0.01)	

Table 4: GBM, fixed 95-point design, all learners at a 20,000-simulation budget with package defaults, scored on 200 test datasets against the exact conjugate posterior. Bias in units of the exact posterior sd (with standard error where it is distinguishable from zero); width is the sd ratio. Inference time is per dataset for 2,000 posterior samples. Timings on one laptop CPU except the Simformer rows (H200 GPU).

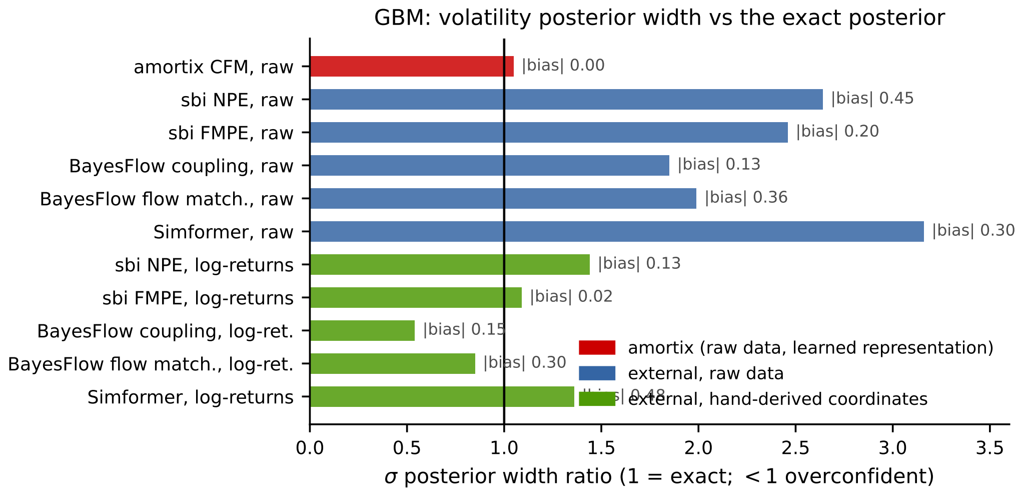

Figure 7: Volatility posterior width (ratio to the exact posterior) for all arms of Table 4. Values below one are overconfident. Annotations give the absolute posterior-mean bias in posterior-sd units.

systems take 8–45 minutes. Inference draws 2,000 posterior samples for a dataset in 426 ms on the laptop CPU and 72 ms (batched; 135 ms single) on the GPU. GPU and CPU runs produce bit-identical posteriors because all stochastic generation is kept on a CPU random number generator by design, so a device change cannot silently alter results. For orientation against non-amortized inference: on GBM, whose likelihood is a three-line formula, adaptive Metropolis costs 42 ms per dataset and amortization buys nothing for a single dataset; on Fisher–KPP, where one likelihood evaluation is one PDE solve (16 ms), the same reference sampler costs minutes per dataset (measured in Table 6); and for Heston or Hodgkin–Huxley no likelihood formula is available at any price. The regime where amortization pays is determined by the likelihood cost, the number of datasets, and — as Table 4 indicates — by whether a trustworthy input representation for a conditional density estimator is available without one.

9 Failure analysis: conditioning of the encoder input

Every accuracy failure encountered during the development of this package — including, we argue, the external failures of Table 4 — reduces to a single cause appearing in three forms: the conditioning of the numbers presented to the set encoder. We document the three forms with their measurements and repairs; all three repairs are learnable modules rather than problem-specific transformations.

Form 1: floating multiplicative scale. On GBM, raw price tokens force the network to divide squared increments by squared levels internally; with global standardization this manifests as a volatility

method (raw values as input)	train	inference	θ: bias / width	σ: bias / width
`amortix` CFM	183 s	32 ms	-0.01 / 1.03	$+0.01$ / 1.05
`sbi` NPE	128 s	10 ms	$+0.07$ / 1.15	$+0.19 \pm 0.19$ / 1.88
`sbi` FMPE	87 s	147 ms	$+0.11$ / 1.01	$+0.08 \pm 0.12$ / 1.60
exact-likelihood MCMC	—	124 ms	reference (two chains per dataset)	

Table 5: Ornstein–Uhlenbeck, fixed 74-point design, 20,000-simulation budget, 100 test datasets, one H200 GPU (MCMC on its CPU; time given for the two validation chains). The additive control: external biases vanish and widths improve but remain inflated; compare Table 4.

method (40 observed values as input)	train	inference	D: bias / width	r: bias / width
`amortix` CFM	379 s	31 ms	-0.05 / 1.02	$+0.13$ / 1.04
`sbi` NPE	218 s	7 ms	$+0.19 \pm 0.12$ / 1.28	-0.06 / 1.26
`sbi` FMPE	115 s	405 ms	$+7.3 \pm 2.7$ / 12.5	-0.25 ± 0.10 / 1.19
PDE-likelihood MCMC	—	271 s	reference (two chains per dataset)	

Table 6: Fisher–KPP, fixed $K = 40$ spatio-temporal design, 20,000-simulation budget, 32 test datasets, one H200 GPU (MCMC on its CPU pool). One likelihood evaluation is one PDE solve, so the reference costs 271 s per dataset; the training cost of the amortized model equals the reference cost of fewer than three datasets.

bias of $+0.48$ posterior sd (confirmed against the closed-form posterior) that no amount of training removes. The repair is the monotone mixture reparametrization of Sec. 4.3, which learns the log coordinate; the Yeo–Johnson ablation there shows why the parametrization, not the hypothesis class, is the operative choice. External estimators exhibit the same form on the same data (Table 4, raw-price rows).

Form 2: cardinality blindness. Normalized attention erases set size, which controls posterior width under design amortization; the probe measurements and the $\log K$/register-token repairs are given in Sec. 4.5, with the arm-by-arm consequences in Table 3.

Form 3: heavy tails. On the Merton jump-diffusion, squared increments have sample standard deviation 3.1×10^4 against a diffusion signal of order 10^{-2}–1; running feature normalization never converges, and the flow trains against a drifting representation, producing a $+1.12$ sd volatility bias. Encoder probing (Sec. 6.3) shows the information is extracted — the failing configuration encodes robust volatility statistics better than the healthy one — so the repair targets representation stability: logarithmic compression of increment features (near-identity on clean diffusions since $\log(1 + x) \approx x$ for small x) restores all 25 evaluation cells. A precision follow-up against the Poisson-mixture reference then isolated a smaller ($+0.27$ sd) residual whose mechanism is a jump-classification threshold that floats with the unknown volatility; a per-dataset set-conditioning module (pooled statistics modulating feature channels, initialized to the identity) removes it as effectively as a hand-crafted per-dataset rescaling ($+0.127 \pm 0.055$ vs $+0.094 \pm 0.054$ sd, statistically indistinguishable). This was one of three such paired comparisons in the project (log coordinate, set size, robust scaling); in all three the learnable module matched the hand-crafted repair once initialized at the working configuration. We summarize the pattern as a design principle: the hand-crafted fix identifies the mechanism and the initialization; the learnable module placed there is what ships.

Three implementation conditions proved necessary for the set-conditioning module and are recorded here because each was found through a failure: the modulation must be zero-initialized (training must start from the working baseline); it must act after feature normalization, not before (the reverse ordering re-creates the representation drift it is meant to cure); and pooled set statistics must exclude padding positions (a masking error here made padded and packed representations of the same dataset inequivalent, detected only because SBC and the exact-reference probe disagreed).

10 Discussion and limitations

The evidence assembled here supports a specific claim: for dynamical-systems inverse problems, the input representation of an amortized posterior estimator is not a preprocessing detail but the dominant accuracy factor, and it can be made part of the model rather than part of the problem specification. The comparison of Sec. 8.5 shows the cost of the alternative — fixed representations with global standardization — across four generative engines, on a problem where the correct answer is known in closed form. The OU control separates the mechanisms: on additive data the bias of the external estimators disappears and their width inflation drops by roughly a factor of two but does not vanish, so the conditioning pathology accounts for the dominant part of the failure and a smaller representation-learning gap accounts for the rest.

Limitations. (i) The remaining inaccuracies of the suite decay as measured power laws in training compute (Fig. 5); we have not searched for methods that improve the exponents. (ii) The power limits of SBC imply that small residuals can only be certified where an exact or validated reference exists; systems without tractable likelihoods inherit the weaker screen, and carrying reference-equipped systems alongside them in one suite is our current mitigation. (iii) The comparison of Sec. 8.5 uses package defaults; it measures the out-of-the-box regime, not the ceiling of any method. (iv) Bayesian experimental design — the intended consumer of design amortization — is not yet demonstrated; the training procedure of Sec. 5 was built so that the required object $p(m \mid S)$ is already coherent across subsets S. (v) No real-data study with posterior predictive checks is included yet. (vi) All priors are uniform on boxes; the probit normalization (2) extends to any factorized prior with tractable CDFs, but correlated priors would require a different normalization.

A Benchmark system definitions

All SDEs are integrated by Euler–Maruyama on the stated fine grid; ODEs by RK4 (Hodgkin–Huxley by a semi-implicit scheme for the gating variables); the PDE by finite differences. Fixed-design observations are token sets over the union of the stated channels; "fast/slow" channels observe every step over a short window and every j-th step over the horizon respectively. Priors are uniform on the stated boxes. Classical estimators (`sota()`) are listed per system.

Linear-Gaussian. $y = Am + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.5^2 I_6)$, with a fixed correlated design matrix $A \in \mathbb{R}^{6\times 4}$; $m \in [-3, 3]^4$. Exact Gaussian posterior; the classical estimator is the posterior mean itself. This system checks the flow engine in isolation from the encoder.

Ornstein–Uhlenbeck. $dX = -\theta X\,dt + \sigma\,dW$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$; $dt = 0.02$, 500 steps; channels: fast (1×50), slow (20×24). Prior $\theta \in [0.3, 3]$, $\sigma \in [0.2, 1.5]$. Classical: exact MLE from per-gap Gaussian transitions.

Geometric Brownian motion. $dS = \mu S\,dt + \sigma S\,dW$, $S_0 = 1$; $dt = 0.01$, 500 steps (five "years"); channels: fast (1×60), slow (10×40). Prior $\mu \in [-0.2, 0.4]$, $\sigma \in [0.1, 0.6]$. Classical: exact MLE on log-returns. Reference: conjugate posterior (Appendix B).

Cox–Ingersoll–Ross. $dX = a(b - X)\,dt + \sigma\sqrt{X}\,dW$; $dt = 0.02$, 500 steps; channels as OU. Prior $a \in [0.3, 3]$, $b \in [0.3, 1.5]$, $\sigma \in [0.1, 0.5]$. Classical: Euler pseudo-likelihood MLE (Cox et al., 1985).

Double well. $dX = (\theta_1 X - \theta_2 X^3)\,dt + \sigma\,dW$; $dt = 0.01$, 1000 steps, noise strong enough for barrier crossings; channels: fast (1×80), slow (25×36). Prior $\theta_{1,2} \in [0.5, 3]$, $\sigma \in [0.4, 1.2]$. Classical: Kramers–Moyal regression.

Stochastic Lotka–Volterra. Prey/predator pair with multiplicative noise, $dx_1 = (\alpha x_1 - \beta x_1 x_2)\,dt + 0.1 x_1\,dW_1$, $dx_2 = (\delta x_1 x_2 - \gamma x_2)\,dt + 0.1 x_2\,dW_2$; both components observed; $dt = 0.01$, 600 steps; channels: fast (1×50), slow (15×40). Prior $\alpha, \gamma \in [0.8, 1.5]$, $\beta, \delta \in [0.4, 1.2]$. Classical: multi-start nonlinear least squares.

FitzHugh–Nagumo. $\dot{v} = v - v^3/3 - w + I + \text{noise}$, $\dot{w} = \epsilon(v + a - bw)$; only v observed at 25 evenly spaced times over horizon 40 ($dt = 0.05$), observation noise 0.05. Prior $a, b \in [0.5, 0.9]$, $\epsilon \in [0.05, 0.3]$, $I \in [0, 0.5]$. Classical: multi-start nonlinear least squares on the deterministic solve.

SEIR with mortality. Five-compartment stochastic epidemic (two-phase contact rate $\beta_1 \to \beta_2$, incubation α, recovery γ_r, mortality γ_d); infected and deceased compartments observed at 20 days over a 60-day horizon ($dt = 0.1$), observation noise 0.004. Prior in Table 7. Classical: nonlinear least squares on (I, D).

system	per-parameter SBC p-values
linear-Gaussian	m_1: 0.06 m_2: 0.60 m_3: 0.38 m_4: 0.56
Ornstein–Uhlenbeck	θ: 0.28 σ: 0.77
SEIR	β_1: 0.33 β_2: 0.22 α: 0.61 γ_r: 0.34 γ_d: 0.53
geometric Brownian motion	μ: 0.26 σ: 0.44
Cox–Ingersoll–Ross	a: 0.65 b: **0.001** σ: 0.73
double well	θ_1: 0.78 θ_2: 0.54 σ: 0.88
stochastic Lotka–Volterra	α: 0.07 β: 0.69 δ: 0.08 γ: 0.35
FitzHugh–Nagumo	a: 0.19 b: 0.15 ϵ: 0.90 I: 0.63
SINDy-SDE	c_0: 0.06 c_1: 0.96 c_2: 0.46 c_3: 0.80 σ: 0.50

Table 7: Per-parameter SBC uniformity p-values for the fixed-design gallery run of Table 2 (500 datasets × 200 draws; $p > 0.05$ passes). The single failure is marked in bold; see Sec. 8.1 for its behavior across retrainings.

SINDy-SDE. $dX = (c_0+c_1X+c_2X^2+c_3X^3)\,dt+\sigma\,dW$ (in the spirit of Brunton et al., 2016); $dt = 0.01$, 1000 steps; channels: fast (1×80), slow (20×45). Prior $c_0, c_2 \in [-0.5, 0.5]$, $c_1 \in [-2, 0]$, $c_3 \in [-1, -0.1]$, $\sigma \in [0.3, 0.8]$. Classical: SINDy-style least squares on Kramers–Moyal estimates.

Heston (variable designs). $dS = \mu S\,dt + \sqrt{v}\,S\,dW_1$, $dv = \kappa(\bar{v} - v)\,dt + \xi\sqrt{v}\,dW_2$, $\mathrm{corr}(dW_1, dW_2) = \rho$ (Heston, 1993); price observed, volatility latent; $dt = 0.01$, 500 steps, designs of $K \in [2, 128]$ random times. Prior $\mu \in [-0.1, 0.3]$, $\kappa \in [0.5, 5]$, $\bar{v} \in [0.02, 0.3]$, $\xi \in [0.1, 1]$, $\rho \in [-0.9, 0]$.

Merton jump-diffusion (variable designs). $dS/S = \mu\,dt + \sigma\,dW + d\big(\sum_i (e^{J_i} - 1)\big)$ with Poisson jump rate λ and $J_i \sim \mathcal{N}(\mu_J, s_J^2)$ (Merton, 1976); $dt = 0.01$, 500 steps, $K \in [2, 128]$. Prior $\mu \in [-0.1, 0.3]$, $\sigma \in [0.1, 0.5]$, $\lambda \in [0.2, 3]$, $\mu_J \in [-0.3, 0.3]$, $s_J \in [0.05, 0.4]$. Reference: Poisson-mixture transition density (Appendix B).

Hénon–Heiles (variable designs). Two-dimensional Hamiltonian oscillator with the coupling potential used in Lubich et al. (2015), frequencies (ω_1, ω_2) and coupling λ; q_1 observed at random times with noise; $dt = 0.05$, 800 steps, $K \in [4, 64]$. Prior $\omega_{1,2} \in [0.7, 1.3]$, $\lambda \in [0.02, 0.22]$.

Hodgkin–Huxley (variable designs). Standard four-dimensional membrane model (Hodgkin and Huxley, 1952); voltage observed at random times; parameters $(g_{\mathrm{Na}}, g_K, g_L, I)$ with prior $[60, 180] \times [10, 50] \times [0.05, 0.5] \times [4, 12]$; 3000 integration steps, $K \in [4, 96]$.

Pharmacokinetics (variable designs). One-compartment oral dosing (Bateman function), concentration $c(t) = \frac{D_0 k_a}{V(k_a - k_e)}\big(e^{-k_e t} - e^{-k_a t}\big)$ with dose $D_0 = 500$; log-normal assay noise (10%); horizon 24 h, $K \in [3, 64]$ irregular draws. Prior $k_a \in [0.5, 4]$, $k_e \in [0.05, 0.4]$, $V \in [20, 100]$. Reference: log-normal likelihood MCMC.

Fisher–KPP (variable designs). $\partial_t\theta = D\,\partial_{xx}\theta + r\,\theta(1 - \theta)$ on $[0, 1]$ with reflecting boundaries, 64-point finite-difference grid, Gaussian initial bump; three fixed sensors at $x = 16/63, 32/63, 48/63$ observed at random (time, sensor) pairs with noise 0.02; $dt = 0.01$, 200 steps, $K \in [4, 96]$. Prior $D \in [0.001, 0.01]$, $r \in [2, 10]$. Reference: Gaussian observation model around the deterministic solve; one likelihood evaluation is one PDE solve.

B Reference posteriors

GBM conjugate posterior. With observations at indices $i_0 = 0 < i_1 < \ldots$ of the fine grid, log-returns $r_k = \log(S_{i_{k+1}}/S_{i_k})$ over gaps τ_k are independent $\mathcal{N}\big((\mu - \sigma^2/2)\tau_k, \sigma^2\tau_k\big)$. In the coordinates $(b, \sigma^2) = (\mu - \sigma^2/2, \sigma^2)$ the model is a Gaussian location–scale family with sufficient statistics $T = \sum_k \tau_k$, $R_1 = \sum_k r_k$, and $SS_w = \sum_k r_k^2/\tau_k - R_1^2/T$; draws from the standard conjugate posterior ($\sigma^2 = SS_w/\chi^2_{n-1}$, $b \mid \sigma^2 \sim \mathcal{N}(R_1/T, \sigma^2/T)$) are transformed to (μ, σ) and importance-resampled to the uniform box prior with the appropriate Jacobian weight. The sampler was validated against exact-likelihood MCMC on the same designs: over 24 comparisons, worst posterior-mean discrepancy 0.099 sd, width ratios 0.91–1.03.

Adaptive Metropolis. Random-walk Metropolis on the box, proposal $x' = x + s\,C\,\xi$, $\xi \sim \mathcal{N}(0, I)$. During burn-in (default $\max(500, n/2)$ iterations) the scalar s is adapted toward a 0.25 acceptance rate in blocks of 50 with a decaying Robbins–Monro gain, and once sufficient burn-in states are collected the shape C is replaced by the Cholesky factor of the empirical covariance rescaled by $2.38/\sqrt{d}$ (Haario et al., 2001); adaptation is frozen for the retained phase. Out-of-box proposals are rejected, which samples the box-truncated target exactly since the proposal is symmetric. Likelihoods: per-gap Euler–Maruyama Gaussian transitions for OU (matching the generative process, plus the stationary density of X_0); the exact GBM log-return density; the Poisson-mixture transition density for Merton (truncated at the numerically negligible jump count); log-normal residuals around the Bateman curve; a Gaussian observation model around the deterministic PDE solve for Fisher–KPP. Every chain used as a reference is paired with a second chain from a well-separated start; between-chain posterior-mean discrepancies are reported alongside the results they certify (Sec. 8.5).

C Correspondence with package options

in this report	in the package
point tokens	`embed="wpoint"`
pair tokens + set conditioning	`embed="wfilm"`
warped values/increments (fixed designs)	`embed="wbasis"` / `"wdiff"`
attention phases from physical time	`rope="time"`
fresh designs each optimizer step	`fit(retokenize=...)` via `make_retokenizer()`
variable-design problem wrapper	`amortix.designs.DesignProblem`
benchmark systems	`amortix.problems` (`DESIGN_ZOO`)
calibration screen / reference machinery	`sbc_design` / `amortix.mcmc` + likelihood factories
comparison scripts (Sec. 8.5)	`examples/baseline_{npe,ou,kpp,bayesflow,simformer}.py`

All defaults named in this report are selected automatically by `FlowPosterior(problem)`; the non-default options exist as reproducible controls for the comparisons above.

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